NNN-B-001194(DSA-DM)
14 June 1966
Interim Revision of
Fed. Spec. DD-B-576
26 May 1942

INTERIM FEDERAL SPECIFICATION

BOTTLES: DROPPER, AND DROPPING TK

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification DD-B-576 dated 26 May 1942.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of glass bottles for dispensing liquids.

1.2 Classification. Bottles covered by this specification shall be of the following types, classes, grades and sizes, as specified (see 6.2).

1.2.1 Types. Type I - Bottle, dropping TK, with ground, grooved-glass stoppers.
Type II - Bottle, dropper, with ground glass pipet dropper and rubber bulb.

1.2.2 Classes. Class 1 - Amber
Class 2 - Clear

1.2.3 Grades. Grade A - Soda lime
Grade B - Borosilicate

1.2.4 Sizes. Size 1 - 30 milliliters (ml.)
Size 2 - 60 milliliters (ml.)
Size 3 -200 milliliters (ml.)

FSC 6640

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2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-102	Preservation, Packaging, and Packing Levels.
FED-STD-123	Marking for Domestic Shipment (Civilian Agencies)

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices, in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS.

ASTM Method C148-59	Polariscopic Examination of Glass Containers.
ASTM Method C225-59	Resistance of Glass Containers to Chemical Attack.

(Copies of ASTM Standards may be obtained upon application to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

U. S. PHARMACOPOEIAL CONVENTION PUBLICATION, INC.

Pharmacopoeia of the United States.

(The Pharmacopoeia of the United States is distributed by the Mack Publishing Company, Easton, Pa.).

3. REQUIREMENTS

3.1 Material.

3.1.1 Glass. Bottles shall be made of high quality chemical-resistant glass. Grade A bottles shall be such that not more than 9.5 ml. of 0.02 normal sulfuric acid shall be needed to neutralize the extract from 10 grams of glass. Grade B bottles shall be such that not more than 1.0 ml of 0.02 normal sulfuric acid shall be needed to neutralize the extract from 10 grams of glass. The solubility for each grade of bottles shall be tested as specified in 4.4.1.

3.1.2 Rubber. The bulb shall be made from an oil-resistant synthetic rubber.

3.2 Construction.

3.2.1 Type I.

3.2.1.1 Bottles. Bottles shall be cylindrically uniform with a flat or slightly concave base and a rounded top (shoulder) finished with a narrow mouth having its inner surface ground to accommodate a ground-in stopper, with diametrically opposed vents for air and liquid. The bottom of the bottle shall rest evenly on a flat surface.

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3.2.1.2 Stopper. A ground stopper shall securely fit the ground surface of the neck of the bottle and shall have a vent for air and a dropper outlet for liquid diametrically opposed. The head of the stopper shall be flat or convex with protruding spout for delivery.

3.2.2 Type II.

3.2.2.1 Bottles. The bottles shall be tapered slightly downward from the shoulder. The bottom of the bottle shall be flat or slightly concave so that it rests evenly on a flat surface. The neck of the bottle shall be taper ground to accommodate the stopper.

3.2.2.2 Pipet. The neck of the pipet shall be taper ground to securely fit the ground surface of the neck of the bottle. The upper end shall be flanged to fit the inner ring in the bulb. The ends of the pipet shall be smoothly finished. When in place, the pipet shall extend to 1/4 inch ± 1/8 inch from the bottom of the bottle.

3.2.2.3 Rubber bulb. The rubber bulb shall be constructed with heavy walls, shall fit tightly around the collar of the pipet and shall be of sufficient drawing strength, when completely compressed and released, to fill the pipet to at least 3/4 of its capacity. Bulb shall be made with an inner ring to fit around the collar of the pipet to prevent the pipet from slipping up into the rubber bulb. Bulb shall be made to fit the unconstricted end of the pipet, so that it shall pass the air test specified in 4.4.2 without showing leaks around the pipet or through defects in the rubber. Bulbs shall pass the accelerated aging test when tested as specified in 4.4.3.

3.3 Dimensions and capacities. Dimensions and capacities shall be as specified in table I.

Table I. Dimensions and Capacities

	Height (inches)				Diameter Bottom of bottle (inches)	Capacity (ml.)
	Type I		Type II			
Size	Overall	Bottle proper	Overall	Bottle Proper		
1	-	-	4-1/8 ± 1/8	2-7/8 ± 1/8	*1-3/8 ± 1/64	30 -0 +5
2	4 ± 1/8	3-5/8 ± 1/8	4-13/16 ± 5/16	3-1/8 ± 1/8	1-5/8 ± 1/8	60 -0 +5
3	5-3/8 ± 1/8	4-7/8 ± 1/8	-	-	2-1/2 ± 1/8	200 -0 +5

* Outside diameter of bottom and body of bottles shall not exceed 1-3/8 inches and when tapered shall be 1-7/16 ± 1/32 inches at the shoulder, so as to properly fit a round receptacle measuring 1-15/32 inches in diameter and 1 inch in height.

3.4 Strain. When tested as specified in 4.4.4, the bottles and pipette droppers shall show no ring strain greater than temper no. 1. Longitudinal strain, if present, shall be faint and highly diffused.

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3.5 Thermal shock. Bottles and stoppers shall show no evidence of corrosion, chipping, scumming, or cracking when tested as specified in 4.4.5.

3.6 Color. Class 1, amber glass, shall be light-resistant, conforming to the requirements of light-resistant containers specified in the Pharmacopoeia of the United States (Edition XVI).

3.7 Identification marking. Each bottle shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.8 Workmanship. Workmanship shall be first class throughout. Bottles shall be properly finished, and free of rough spots, bubbles, and blisters which will detract from their appearance or impair their strength or serviceability.

4. SAMPLING, INSPECTION AND TEST PROCEDURES.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination(such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-3	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

Table II. Test Methods

Component Unit of Product	Characteristic	Requirement	Test Procedure	Inspection level	AQL
Bottle and Stopper	Solubility	3.1.1	4.4.1	S-2	*
Bulb and Stopper	Air Test	3.2.2.3	4.4.2	S-2	*
Bulb	Aging	3.2.2.3	4.4.3	S-2	*
Bottle and Stopper	Strain	3.4	4.4.4	S-2	*
Bottle and Stopper	Resistance to shock	3.5	4.4.5	S-2	*
Bottle and Stopper	Fill	3.2.2.2	4.4.6	S-2	*

* Acceptance number shall be zero.

4.3 Examination. The bottles shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects *

Major
- 101 Bottle and stopper not free of cracks, chips, or breaks.
- 102 Color of class 1 bottles not as specified.
- 103 Bottle not stable on a smooth, flat surface.
- 104 Top (shoulders) of bottle not rounded.
- 105 Identification marking illegible, incorrect or incomplete.

Minor
- 201 Bottles and stoppers not free of rough spots, bubbles and blisters.
- 202 Pipet, as required, not smoothly finished.

*Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The bottles shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. All applicable tests shall be conducted as specified.

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4.4.1 Solubility. The solubility test shall be conducted in accordance with ASTM Method C-225-59, test method P-W except the oven shall be maintained at a temperature of 90 degrees Centigrade.

4.4.2 Air test. The rubber bulb, when distended under 20 pounds air pressure, shall show no leaks around the pipet, or other defects in the rubber.

4.4.3 Accelerated aging. Test bulbs shall be subjected to the action of oxygen under pressure of 300 pounds per square inch, ± 10 pounds, and at a temperature of 70° C. ± 1° C, for 3 days. After 24 hours at room temperature, there shall be no change in the condition of the bulbs that might affect their serviceability.

4.4.4 Strain. The bottle and stopper shall be tested for strain in accordance with Method A of ASTM Designation C148.

4.4.5 Resistance to shock. Bottles and stoppers, with stoppers not in place, shall be immersed in tap water at 18.5° C ± 2.5° C (65° F ± 5° F), allowed to remain 10 minutes, then transferred suddenly into boiling water. This operation shall be repeated five times.

4.4.6 Fill (Type II). The assembled dropper and bulb shall be tested using water, to determine the capacity of the dropper, and the amount of water drawn into the pipet by completely compressing the bulb.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One bottle, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table III shall be packaged in each unit, intermediate, and exterior container, as applicable for the required level of protection specified in the procurement document.

Table III. Packaging Quantities

Size	Unit package	Intermediate package	Exterior container
1	NOT REQUIRED	12 units	288 units
2		12 units	144 units
3		12 units	24 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit preparation. A strip of paper shall be interposed between the glass stopper and the neck of the bottle.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-676, except that commercial colors shall be acceptable, or PPP-B-636. Carton design shall include full height partitions, half-slotted style. The partitions shall form an individual, snug fitting cell for each bottle. Partitions shall be fabricated of material equal to the strength of board used for fabrication of the carton.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2. Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Intermediate packages. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bottles covered by this specification are intended for use in dispensing biological stains, reagents and indicator solutions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Type, class, grade, and size required (see 1.2)
(c) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes,grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD, MU, MO
Navy - MS, SH
Air Force - 03, 82

Civilian Agency Interest:
GSA-FSS
HEW-PHS
COM-NBS

Preparing activity:
Defense Supply Agency -DM

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

NNN-B-001194(DSA-DM)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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